Lesson 1-1 • Limits to a Point on a Graph

Notes (filled in)

Objective: find left-hand, right-hand and two-sided limits from a graph.

Limit notation

From the left

$\lim_{x \to c^-} f(x)$

x approaches c with values less than c

From the right

$\lim_{x \to c^+} f(x)$

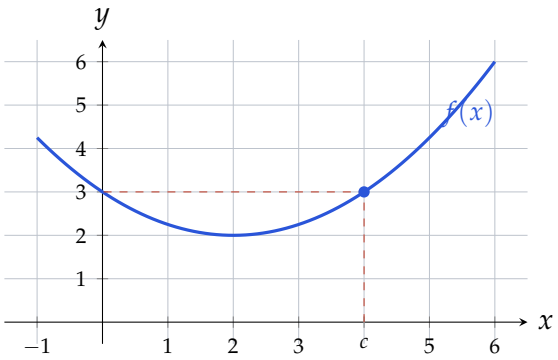
x approaches c with values greater than c

From both sides

$\lim_{x \to c} f(x)$

both one-sided limits are equal

My first limits



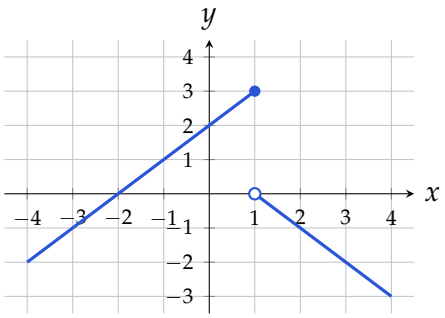
Use the graph, with $c = 4$.

1) $\lim_{x \to 4^-} f(x) =$ 3 2) $\lim_{x \to 4^+} f(x) =$ 3 3) $\lim_{x \to 4} f(x) =$ 3

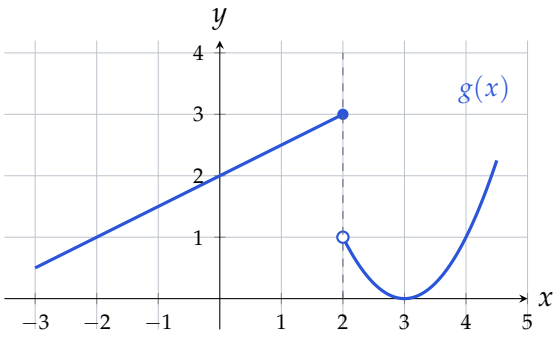
A limit is a y-value (a height): the height the graph approaches as x gets close to c.

Challenge

Draw the graph of a function whose left-hand and right-hand limits at $x = 1$ are **not** the same.



Example: the left piece arrives at height 3, the right piece starts at height 0, so $\lim_{x \to 1} f(x)$ does not exist.

Example: one-sided limits are not the same	 1) $\lim_{x\to 2^-} g(x) =$ 3 2) $\lim_{x\to 2^+} g(x) =$ 1 3) $\lim_{x\to 2} g(x) =$ DNE 4) $g(2) =$ 3 (the height *at* $x = 2$: the filled dot) If the left and right limits are different, the two-sided limit does not exist (DNE).
Summary	A limit is the height the graph approaches. $\lim_{x\to c} f(x)$ exists only when the left and right limits are equal. The value $f(c)$ is the height of the filled dot at c; it can be different from the limit, or not exist.